Gesture recognition

Gesture recognition

1. Gesture recognition
2. Preview of the effect
    2.1. The USB camera
    2.2. The CSI camera

MediaPipe is a cross-platform machine learning framework that can process key point detection such as hands, faces, and poses in videos and images in real time.

1. Gesture recognition

MediaPipe gesture recognition can detect and classify gestures: it can recognize the Zero, One, Two, Three, Four, Five, Six, Seven, Eight, Ok, Rock, Thumb_up (thumbs up), Thumb_down (thumbs down), and Heart_single (single-hand heart) gestures.

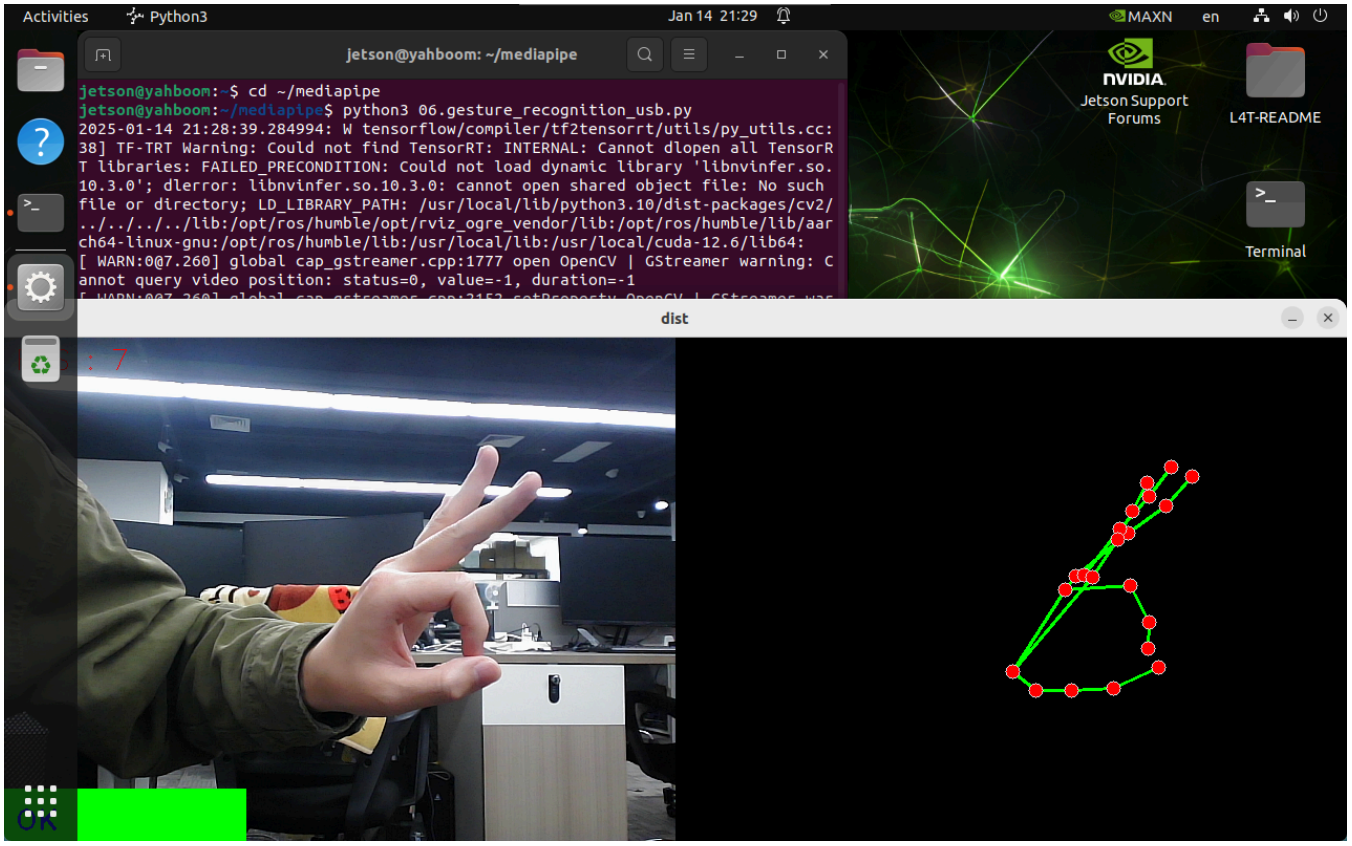
2. Preview of the effect

2.1. The USB camera

```
cd ~/mediapipe
```

```
python3 06.gesture_recognition_usb.py
```

Note: click the camera window and press `q` to exit the program!

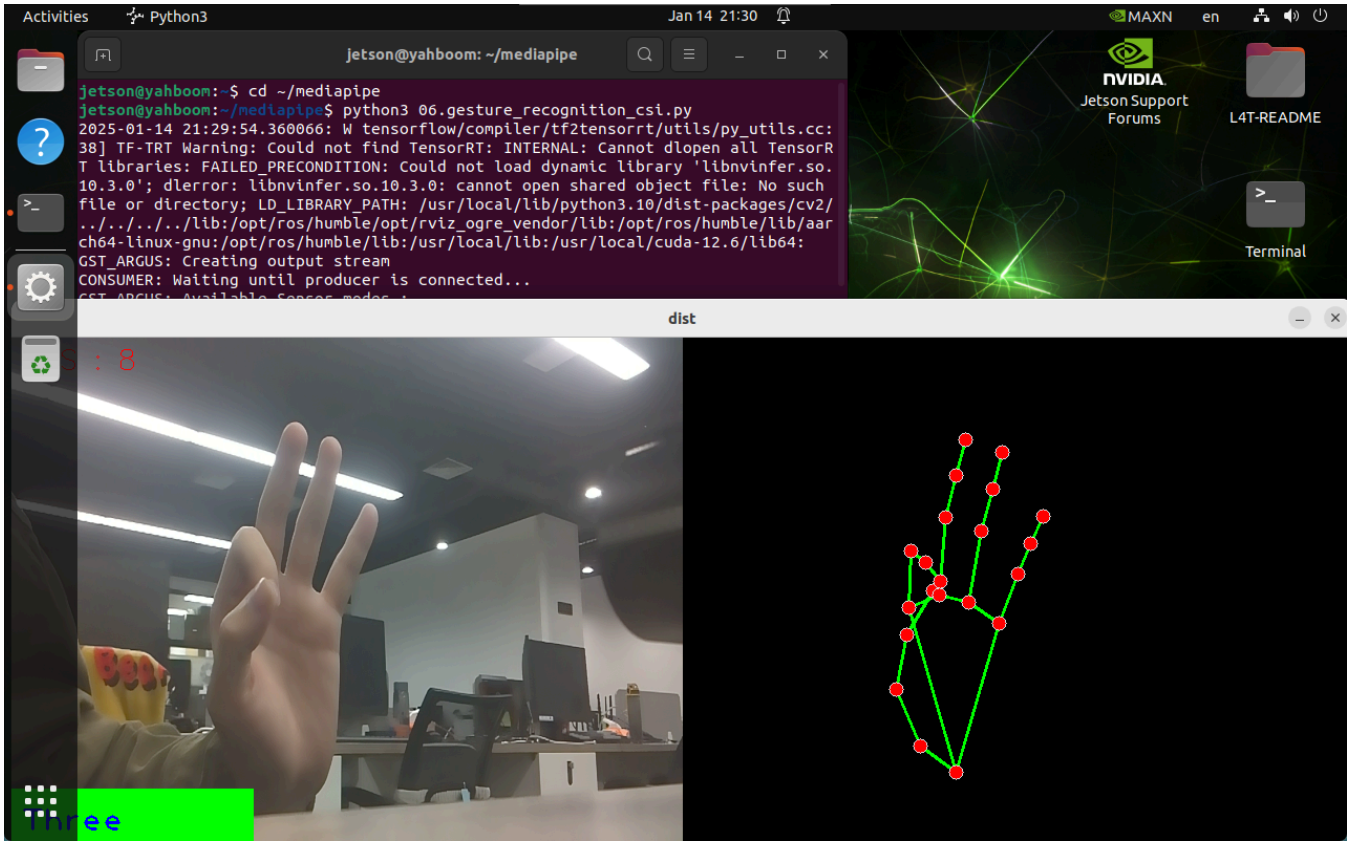


2.2. The CSI camera

```
cd ~/mediapipe
```

```
python3 06.gesture_recognition_csi.py
```

Note: click the camera window and press `q` to exit the program!


Activities
Python3
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jetson@yahboom: ~/mediapipe
jetson@yahboom:~$ cd ~/mediapipe
jetson@yahboom:~/mediapipe$ python3 06.gesture_recognition_csi.py
2025-01-14 21:29:54.360066: W tensorflow/compiler/tf2tensorrt/utils/py_utils.cc:38] TF-TRT Warning: Could not find TensorRT: INTERNAL: Cannot dlopen all TensorRT libraries: FAILED_PRECONDITION: Could not load dynamic library 'libnvinfer.so.10.3.0'; dlerror: libnvinfer.so.10.3.0: cannot open shared object file: No such file or directory; LD_LIBRARY_PATH: /usr/local/lib/python3.10/dist-packages/cv2/../../../../lib:/opt/ros/humble/opt/rviz_ogre_vendor/lib:/opt/ros/humble/lib/aarch64-linux-gnu:/opt/ros/humble/lib:/usr/local/lib:/usr/local/cuda-12.6/lib64:
GST_ARGUS: Creating output stream
CONSUMER: Waiting until producer is connected...
NVIDIA
Jetson Support Forums
L4T-README
Terminal
dist
three